REVENUE OF DIFFERENT LOAN TYPE & ANALYZE THE DEFAULTER RATIO (2024-2025)

1.PROJECT OBJECTIVE:

- The primary objective of this project is to analyse multi bank loan data across the different locations and financial years to identify trends,

- This project involves cleaning, transforming and analysing using **EXCEL** and creating an interactive **POWER BI DASHBOARD** to deliver meaningful business insights.

2.DATA SOURCES:

- **Sources description and timeline:** External source & Time period (2014-2015)
- **Domain:** Banking sector (Assets)

3.PROBLEM STATEMENT

- Banks are disbursed different types of loans to the customer across the countries so we have to find Revenue of loan given by bank
- To analyse the Average rate of interest of each loan segment
- Banks revenue depends upon the rate of interest and loan tenure
- Finally find out the DEFAULTER ratio, what is the reason behind that

4.ATTRIBUTE (COLOUMN/FEATURES) DETAILS:

Attribute Name	**Data type**	**Description**
Bank Name	Categorial	Name of the different bank
Branch Id	Integer/string	Unique id for identify the branch
Customer Name	String (Text)	Name of the customer
Country	String (Text)	Country name
City	String (Text)	City name
Loan Type	Categorial	Categories of loan type
Customer Segment	Categorial	Categories of customer segment
Loan Amount	Numeric (Integer)	Loan amount availed by customer

Interest Rate	Numeric (Integer)	Rate of interest to the respective loan
Tenure Year	Numeric (Integer)	Loan tenure
Credit Score	Numeric (Integer)	Credit score to the respective customer
Loan Status	Categorial	Categories of loan status
Estimated Revenue	Numeric (Integer)	Revenue amount towards loan amount
Financial Year	Numeric (Integer)	Loan taken year

5.TOOL & TECHNOLOGIES

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6.DATA PRE-PROCESSING (EXCEL/POWER QUERY)

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, Standardized formats (UPPER CASE), Trim, Cleaning using **POWER QUERY** Mode
- Using RIGHT function to take year separately
- Using DATA BARS (conational formatting) to view the range of revenue amount
- Values changing general to CURRENCY type
- Using DESCRIPTIVE ANALYSIS (AI TOOLPAK) to view the mean, median, standard deviation, minimum, maximum, sum, count, etc….
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.

7.KEYINSIGHTS

BANK vs LOAN AMOUNT

Row Labels	Sum of LOAN_AMOUNT
EVERGREEN BANK LTD	₹ 2,57,79,353.00
FIRST ALLIANCE BANK	₹ 2,15,24,132.00
GLOBAL TRUST BANK	₹ 2,71,32,452.00
LIBERTY BANKING GROUP	₹ 2,53,81,097.00
METROPOLITAN BANK CORP	₹ 2,45,54,435.00
NATIONAL CAPITAL BANK	₹ 2,53,59,728.00
PIONEER CREDIT BANK	₹ 2,45,80,403.00
PREMIER FINANCIAL BANK	₹ 2,41,89,580.00
SUMMIT NATIONAL BANK	₹ 1,76,43,774.00
UNITED COMMERCIAL BANK	₹ 1,91,90,858.00
Grand Total	**₹ 23,53,35,812.00**

BANK vs REVENUE

Row Labels	Sum of ESTIMATED_REVENUE
EVERGREEN BANK LTD	₹ 17,86,124.00
FIRST ALLIANCE BANK	₹ 17,09,675.00
GLOBAL TRUST BANK	₹ 17,92,007.00
LIBERTY BANKING GROUP	₹ 18,27,822.00
METROPOLITAN BANK CORP	₹ 18,10,472.00
NATIONAL CAPITAL BANK	₹ 17,38,617.00
PIONEER CREDIT BANK	₹ 18,79,666.00
PREMIER FINANCIAL BANK	₹ 16,51,529.00
SUMMIT NATIONAL BANK	₹ 12,70,312.00
UNITED COMMERCIAL BANK	₹ 14,93,991.00
Grand Total	**₹ 1,69,60,215.00**

- High loan disbursement-**Global trust bank**
- High revenue -**Pioneer credit bank**

LOAN TYPE vs PERCENTAGE OF REVENUE

Row Labels	Sum of ESTIMATED_REVENUE	Sum of REVENUE (%)
HOME LOAN	₹ 53,26,812.00	11.08450541
OVERDRAFT	₹ 30,53,651.00	31.29938552
PERSONAL LOAN	₹ 11,96,411.00	23.32129278
PROPERTY LOAN	₹ 61,67,990.00	11.3111816
VEHICLE LOAN	₹ 12,15,351.00	14.74270533
Grand Total	**₹ 1,69,60,215.00**	**91.75907064**

LOAN vs TENURE(avg)

Row Labels	Average of TENURE_YEAR
HOME LOAN	20.72
OVERDRAFT	1.91
PERSONAL LOAN	2.98
PROPERTY LOAN	19.60
VEHICLE LOAN	5.02
Grand Total	**9.99**

- **Overdraft** and **Personal** loans are highest percentage of revenue product to the bank due to these two products having low average tenure(loan repayment period)

LOAN WISE TOTAL AMOUNT & REVENUE IN LAST TWO FINANCIAL YEAR

	Column Labels					
	₹ 2,024.00		₹ 2,025.00		Total Sum of LOAN_A	Total Sum of ESTIMAT
Row Labels	Sum of LOAN_AMOU	Sum of ESTIMATED_R	Sum of LOAN_AMOU	Sum of ESTIMATED_REVENUE		
HOME LOAN	₹ 3,46,07,639.00	₹ 21,53,737.00	₹ 5,42,26,972.00	₹ 31,73,075.00	₹ 8,88,34,611.00	₹ 53,26,812.00
OVERDRAFT	₹ 78,79,168.00	₹ 12,99,428.00	₹ 1,09,48,061.00	₹ 17,54,223.00	₹ 1,88,27,229.00	₹ 30,53,651.00
PERSONAL LOAN	₹ 44,59,692.00	₹ 5,69,248.00	₹ 45,79,821.00	₹ 6,27,163.00	₹ 90,39,513.00	₹ 11,96,411.00
PROPERTY LOAN	₹ 6,02,36,105.00	₹ 35,76,518.00	₹ 4,19,36,816.00	₹ 25,91,472.00	₹ 10,21,72,921.00	₹ 61,67,990.00
VEHICLE LOAN	₹ 86,17,054.00	₹ 6,34,278.00	₹ 78,44,484.00	₹ 5,81,073.00	₹ 1,64,61,538.00	₹ 12,15,351.00
Grand Total	**₹ 11,57,99,658.00**	**₹ 82,33,209.00**	**₹ 11,95,36,154.00**	**₹ 87,27,006.00**	**₹ 23,53,35,812.00**	**₹ 1,69,60,215.00**

- Compared to last FY(2024),current FY(2025) Generate more revenue across all loan product

LOAN vs LOAN STATUS				
Count of LOAN_STA	Column L			
Row Labels	ACTIVE	CLOSED	DEFAULTED	Grand Total
HOME LOAN	49	61	72	182
OVERDRAFT	63	53	75	191
PERSONAL LOAN	67	57	52	176
PROPERTY LOAN	71	57	61	189
VEHICLE LOAN	72	74	54	200
Grand Total	**322**	**302**	**314**	**938**

- **In this pivot table, Overdraft and home loans are having more number of defaulters**

  REASON

- **Home loan – Loan tenure is higher compared to other loans**
- **Overdraft loan need to renewal year on year and having more charges compared to other loan, difficult to maintain**

POWER BI (DATA VISUALIZATIONS)

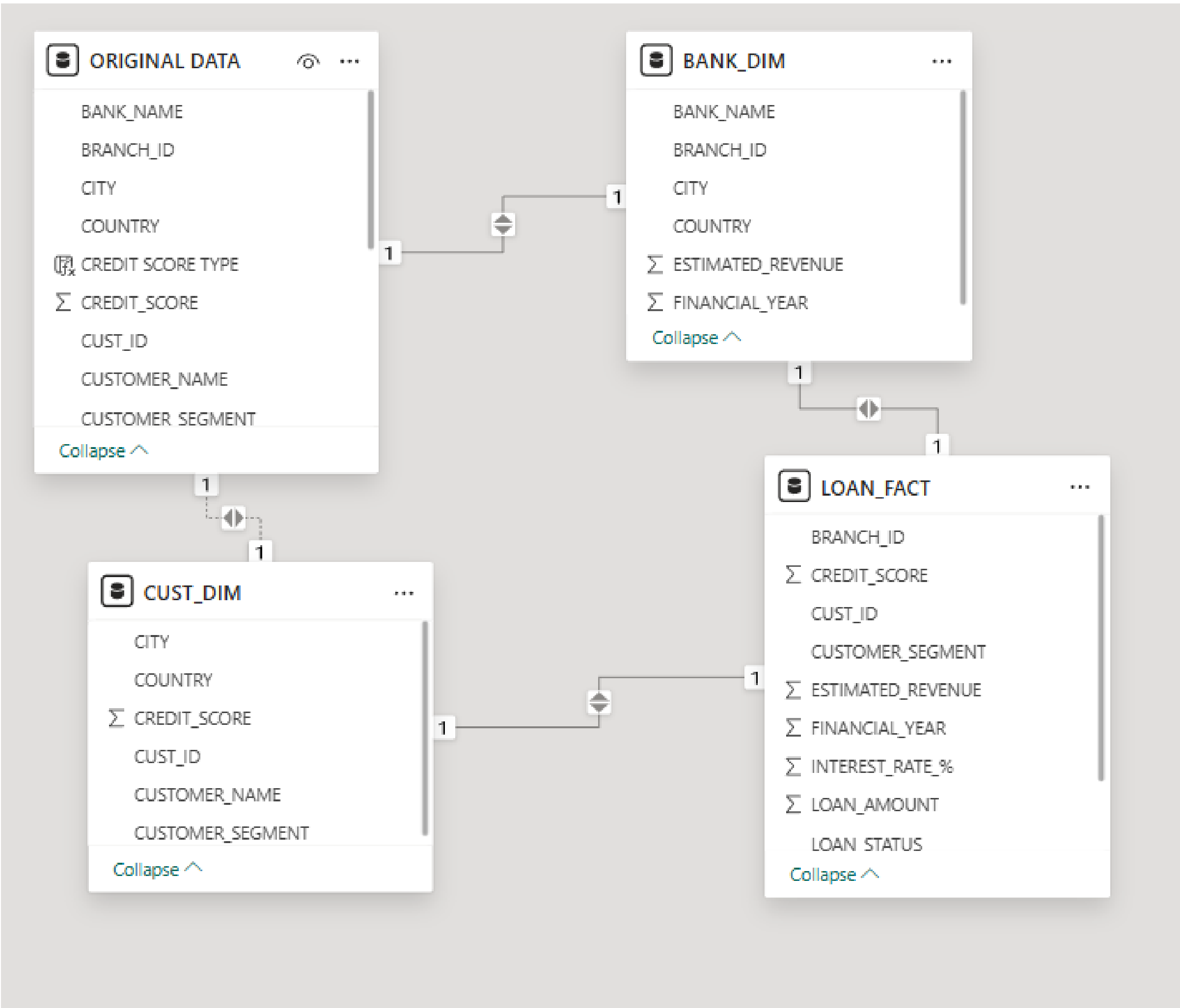
1.OBJECTIVE

Power BI is creating an interactive dashboard to deliver meaningful business insights

- To analysis bank wise total loan amount disbursement and revenue generate
- Highest revenue generates loan product to the bank
- To find the average tenure of each loan
- To analysis of loan status of every customer

2.DATA MODELING

- Connecting tables using relationship
- Creating fact table and dimension table



3.CREATE COLOUMN & DAX FORMULA

CREATE COLOUMN:

CREDIT SCORE TYPE = IF('ORIGINAL DATA'[CREDIT_SCORE]>600,"GOOD",IF('ORIGINAL DATA'[CREDIT_SCORE]<=500,"BELOW","AVERAGE"))

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1 CREDIT SCORE TYPE = IF('ORIGINAL DATA'[CREDIT_SCORE]>600,"GOOD",IF('ORIGINAL DATA'[CREDIT_SCORE]<=500,"BELOW","AVERAGE"))
```

_SEGMENT	LOAN_AMOUNT	INTEREST_RATE_%	TENURE_YEAR	CREDIT_SCORE	LOAN_STATUS	ESTIMATED_REVENUE	FINANCIAL_YEAR	REVENUE (%)	CREDIT SCORE TYPE
	2099	20	3	537	CLOSED	415	2025	0.197713196760362	AVERAGE
	71507	5	3	412	DEFAULTED	3869	2024	0.0541065909631225	BELOW
	149476	11	3	765	CLOSED	16084	2025	0.107602558270224	GOOD
	136162	4	3	662	ACTIVE	5882	2025	0.0431985429121194	GOOD
	55091	5	3	801	CLOSED	3019	2024	0.054800239603565	GOOD
	199871	17	3	516	ACTIVE	33319	2024	0.166702523127417	AVERAGE
	112343	5	3	846	DEFAULTED	6010	2025	0.0534968800904373	GOOD
	194308	12	3	473	DEFAULTED	23647	2025	0.121698540461535	BELOW
	18836	19	3	539	DEFAULTED	3505	2024	0.186079847101295	AVERAGE
	198834	18	3	798	ACTIVE	35671	2024	0.179400907289498	GOOD
	60984	14	3	372	ACTIVE	8428	2024	0.138200183654729	BELOW
	140350	17	3	383	CLOSED	23902	2025	0.17030281439259	BELOW
	172168	19	3	442	DEFAULTED	32144	2025	0.186701361460899	BELOW
	23476	11	3	382	DEFAULTED	2594	2024	0.110495825523939	BELOW
	172487	17	3	634	ACTIVE	29461	2024	0.170801277777456	GOOD
	114810	6	3	845	ACTIVE	7141	2025	0.0621984147722324	GOOD
	14733	14	3	584	ACTIVE	2011	2024	0.136496300821286	AVERAGE
	78139	10	3	307	DEFAULTED	7853	2024	0.100500390330053	BELOW
	109825	12	3	832	DEFAULTED	13355	2025	0.121602549510585	GOOD
	60201	9	3	567	ACTIVE	5177	2024	0.0859952492483514	AVERAGE
	118890	11	3	548	DEFAULTED	12555	2025	0.10560181680545	AVERAGE
	68670	12	3	537	CLOSED	8584	2024	0.125003640599971	AVERAGE
	60294	16	3	453	CLOSED	9816	2024	0.162802268882476	BELOW
	67189	13	3	741	DEFAULTED	8889	2025	0.132298441709208	GOOD
	57854	6	3	348	CLOSED	3367	2024	0.0581982231133543	BELOW

- To find the credit score variance using add coloumn option

DAX FORMULAS (USING FOR THIS ANALYSIS)

I. TOTAL LOAN = SUM('ORIGINAL DATA'[LOAN_AMOUNT])
   Total loan = 235335812

II. TOTAL REVENUE = SUM('ORIGINAL DATA'[ESTIMATED_REVENUE])

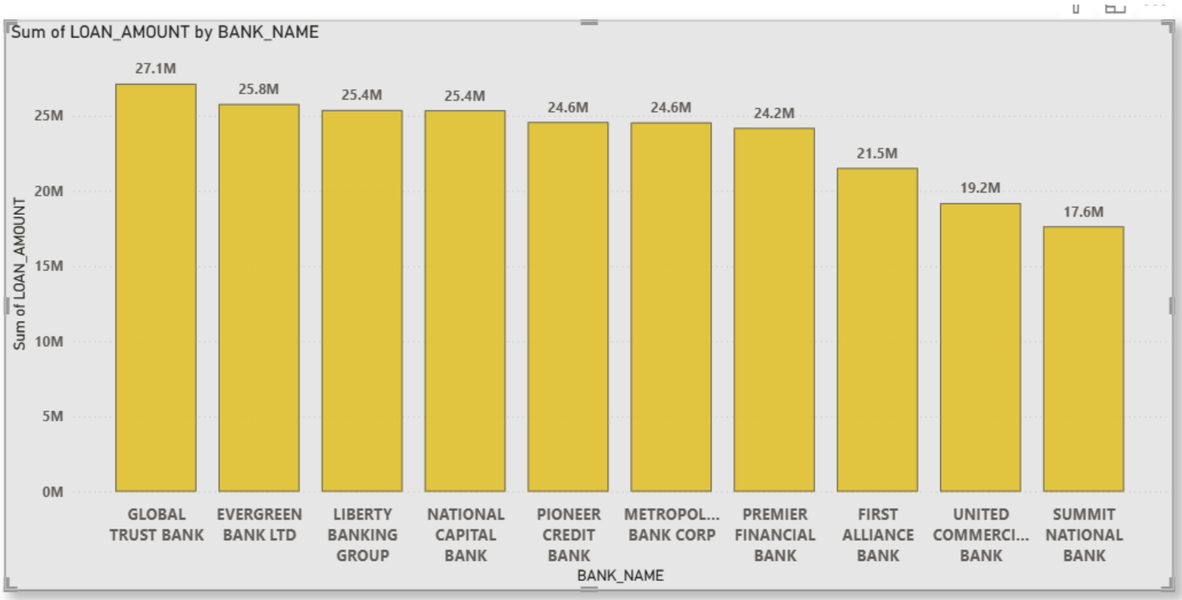
   Total revenue = 16960215

III. AVG CREDIT SCORE = AVERAGE('ORIGINAL DATA'[CREDIT_SCORE])

   Avg credit score = 578.68

4.ANALYSE AND VISUALIZATION

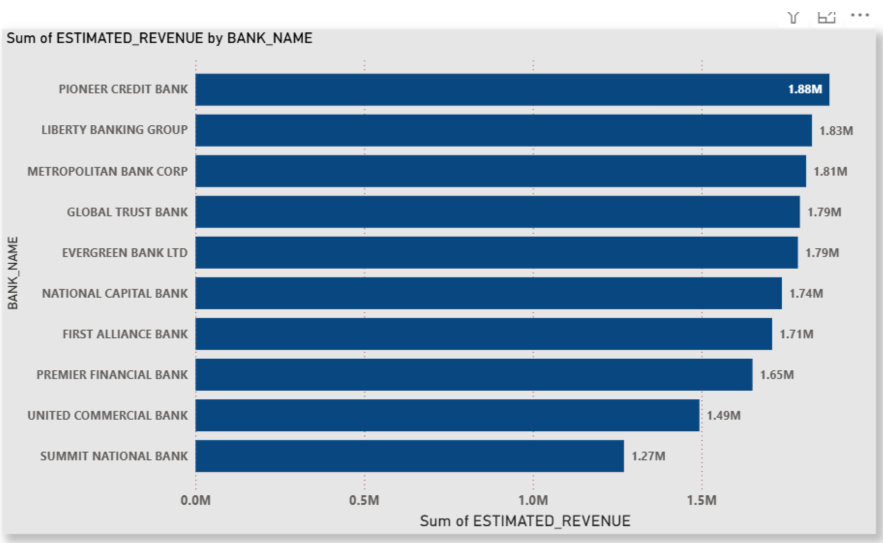
BANK WISE HIGHEST LOAN DISBURSEMENT:



Insights:

- Most banks are clustered between **24M – 26M**, showing fairly balanced competition
- The gap between highest and lowest bank is around **9.5M**
- High performing banks are likely having strong **customer base, aggressive loan product, better market penetration**
- Low performing banks may need **improved loan schemes, better interest rate strategy, increased digital reach**

BANK WISE HIGHEST REVENUE:



Insights:

- Some banks with high loan amounts are not No.1 in revenue, this suggests interest rate difference or margin difference

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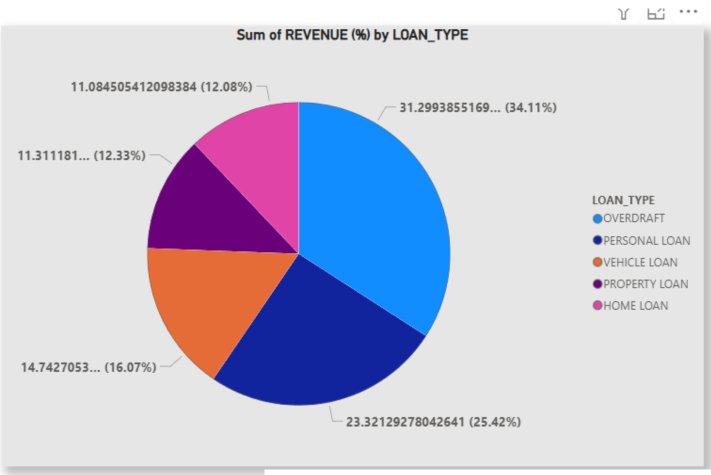
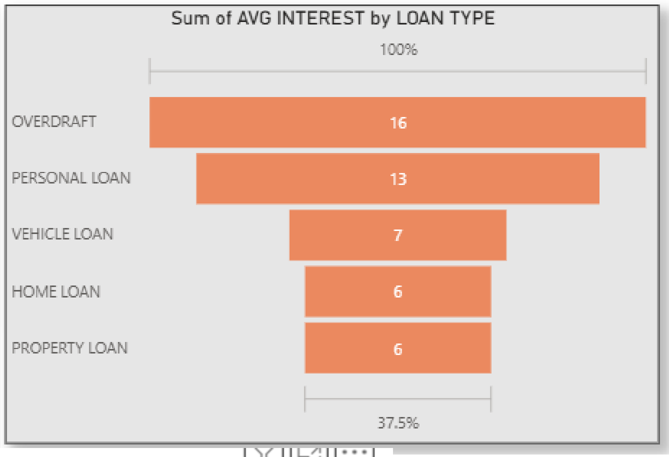
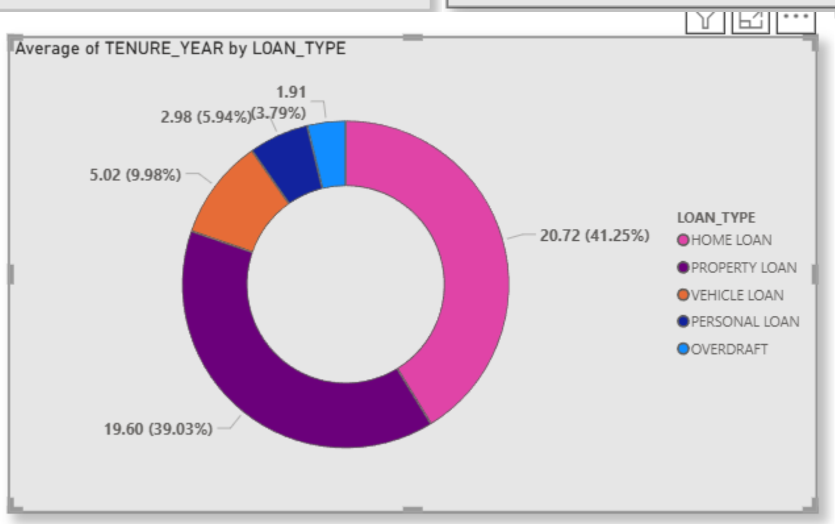
REVENUE CONTRIBUTION BY LOAN TYPES:







Insights:

- **OD** contribute the largest share of revenue, likely higher rate of interest and short-term charges
- **Personal loan** second highest revenue generator, high demand retail product
- **Property loan** and **home loan** is the lowest revenue generate product due to lowest interest rate and long tenure

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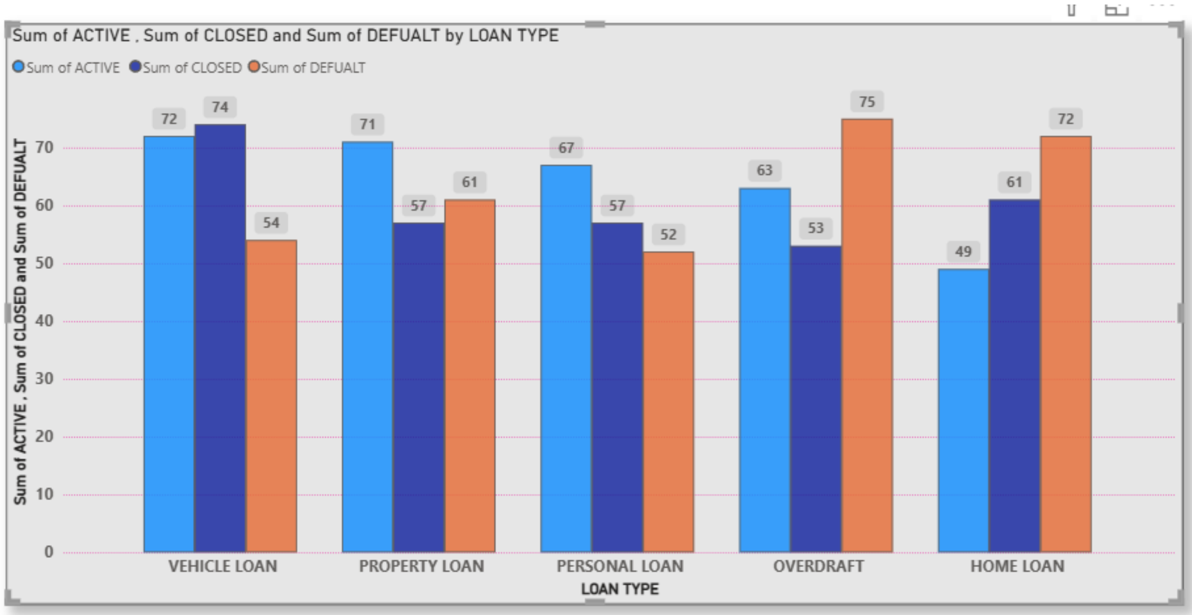
LOAN STATUS ANALYSIS BY LOAN TYPE:



Insights:

- **Vehicle loan** has high closed status, strong repayment performance, healthy loan portfolio
- **Property loan** risk level is moderate, need tights credit checks
- **Personal loan** has lowest defaults, best risk adjusted product
- **OD** having very high default rate – 75, risk product
- Even though revenue % was higher earlier (34%), default is also highest
- **Home loan** second highest default rate

Important observation: High revenue product = High risk product

5.TYPES OF ANLYSIS:

DESCRIPTIVE ANALYSIS:

In descriptive analysis we used bank wise highest loan disbursement details

- Total loan disbursement across banks ranges between 17.6M and 27.1M showing relatively balanced competition
- The market is competitive, with only moderate variation in total distribution across banks

DIAGNOSTIC ANALYSIS:

- Od is usually short-term loan, unsecured, higher interest rate
- Customer with weaker credit profile may use of it
- Risk based pricing is high – high revenue

- But repayment discipline is lower- higher default

PREDICTIVE ANALYSIS:

LOAN TYPE	RISK LEVEL	PERFORMENCE
Vehicle loan	Low	Strong
Personal loan	Low	Strong
Property loan	Medium	Moderate
OD	High	Risk
Home loan	High	Risk

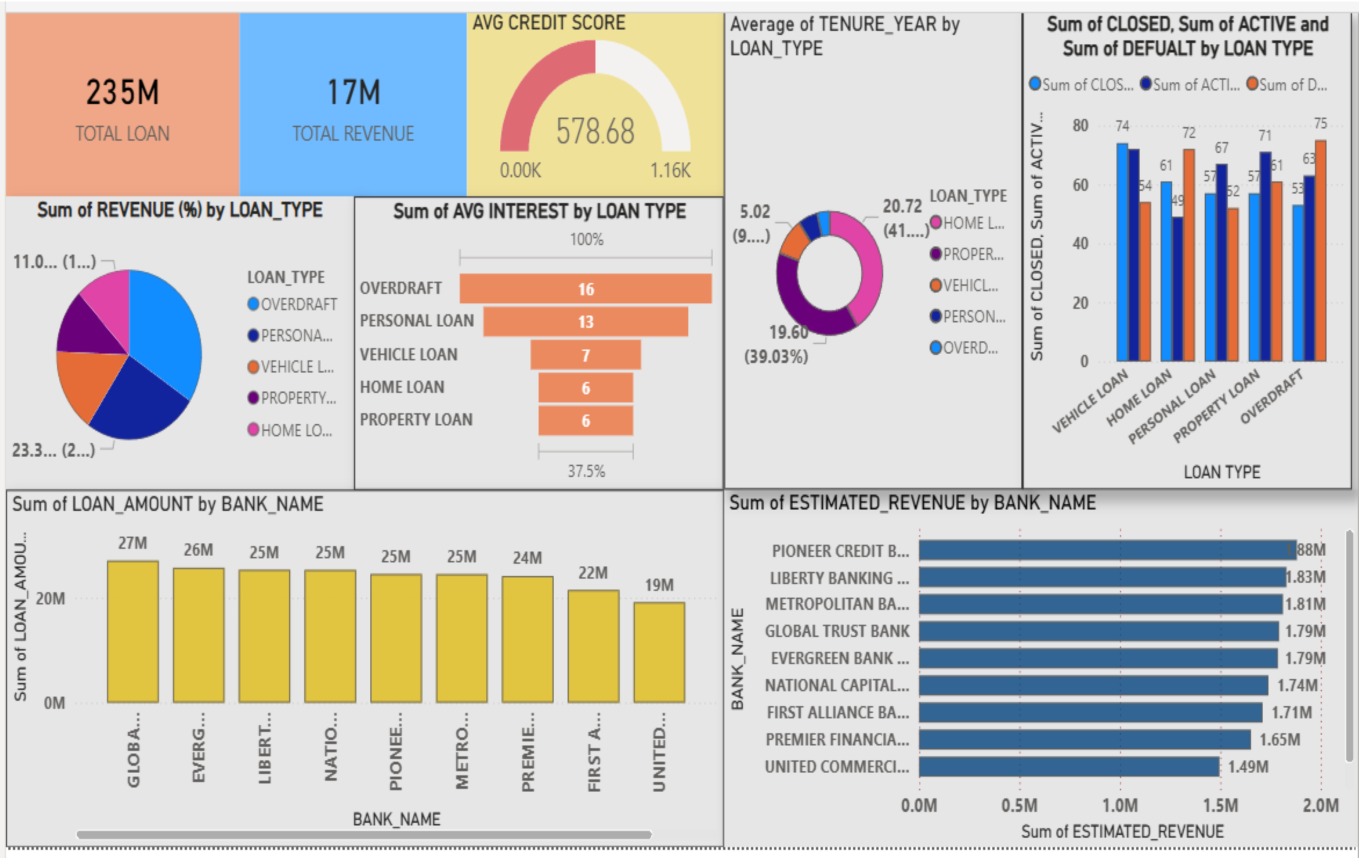
PRESCRIPTIVE ANALYSIS:

- Apply strict credit score threshold
- Reduce OD limit to risky customers
- Introduce income verification
- Increase interest rate for high-risk segments
- Offer low interest for low-risk customers
- Monitoring delay payments
- Trigger reminders

Reduce depency from 34% to safer 25-28%

Goal – Maintain revenue but reduce default rate

6.DASHBOARD:


235M
TOTAL LOAN
17M
TOTAL REVENUE
AVG CREDIT SCORE
578.68
0.00K
1.16K
Average of TENURE_YEAR by LOAN_TYPE
Sum of CLOSED, Sum of ACTIVE and Sum of DEFUALT by LOAN TYPE
Sum of REVENUE (%) by LOAN_TYPE
Sum of AVG INTEREST by LOAN TYPE
OVERDRAFT 16
PERSONAL LOAN 13
VEHICLE LOAN 7
HOME LOAN 6
PROPERTY LOAN 6
Sum of LOAN_AMOUNT by BANK_NAME
Sum of ESTIMATED_REVENUE by BANK_NAME
PIONEER CREDIT B... 1.88M
LIBERTY BANKING ... 1.83M
METROPOLITAN BA... 1.81M
GLOBAL TRUST BANK 1.79M
EVERGREEN BANK ... 1.79M
NATIONAL CAPITAL... 1.74M
FIRST ALLIANCE BA... 1.71M
PREMIER FINANCIA... 1.65M
UNITED COMMERCI... 1.49M